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Video Game Sales Prediction

Overview

For this project I looked at video game sales data and built a model to predict how well a game would sell globally. The dataset I used is the Video Game Sales dataset from Kaggle (https://www.kaggle.com/datasets/gregorut/videogamesales), which has sales figures for thousands of games across different platforms and genres. Since I'm predicting a continuous number, global sales in millions, this is a supervised regression problem. I used a Decision Tree Regressor to build the model.

Dataset Description

The dataset contains sales information for video games that sold over 100,000 copies. It has 16,324 rows and 11 columns. The main features are the game's platform, genre, year of release, and publisher. There are also sales columns broken down by region: North America, Europe, Japan, and other regions, and then a global sales column, which is what I used as the target variable. There were 36 missing values in the Publisher column, which I dropped since it was a small enough amount that it wouldn't really affect the data much.

Modeling Approach

Before training the model, I dropped the Name and Rank columns since they don't really tell you anything useful about how a game will sell. I also dropped the regional sales columns because global sales is basically just the sum of all of them, so keeping them in would just be giving the model the answer. After that, I used one-hot encoding on the Platform, Genre, and Publisher columns to convert them into numbers since the model can't work with text. I ended up with 620 features after encoding.

I chose a Decision Tree Regressor because it's straightforward and works well with a mix of categorical and numeric features without needing any scaling. I also set a max depth of 6 to keep the tree from overfitting to the training data, which was a problem when I first ran it without any depth limit. I used RMSE and R2 as my evaluation metrics because this is a regression problem. RMSE gives you the average error in the same units as the target variable, which is millions of sales, and R2 tells you how much of the variation in sales the model is actually explaining (James et al., 2013).

Results

After training on 80% of the data and testing on the remaining 20%, the model got an RMSE of 1.894 and an R2 of 0.0923. The RMSE means the model's predictions are off by about 1.89 million units on average. The R2 of 0.0923 means the model is only explaining about 9% of the variation in global sales, which honestly isn't great but isn't surprising given how unpredictable video game sales can be.

The actual vs predicted plot from the notebook shows most games clustered near zero, which makes sense because the majority of games in the dataset sold under 1 million copies. The model really struggled with the outliers, the big hit titles that sold way more than anything else in the dataset, which pulled the error up and kept R2 low.

Interpretation for a Non-Technical Audience

Basically what this model is trying to do is look at information about a video game, like what console it's on, what genre it is, and who made it, and use that to guess how many copies it will sell worldwide. It’s like trying to predict how well a movie will do at the box office before it comes out, just based on things like the studio, the genre, and what time of year it releases.

The model got some things right but struggled a lot with the really big hits. Most video games sell a fairly small number of copies, so the model got decent at predicting those, but when a game like RDR2 or GTA comes along and sells tens of millions of copies, the model really had no way to see that coming just from the features we gave it. So while it's a decent starting point, you wouldn't want to use this to actually make business decisions about which games to publish.

Limitations and Potential Bias

One clear limitation is that the model has a pretty low R2 score, meaning it's not capturing most of what drives video game sales. Things like marketing budget, review scores, and whether a game had a big franchise behind it aren't in this dataset at all, and those probably matter a lot more than platform or genre alone.

There's also some potential bias in the dataset itself. It only includes games that sold over 100,000 copies, so the model never learned anything about games that sold poorly. That means it's probably skewed toward predicting higher sales than it should for unknown titles. On top of that, the dataset is heavier on older games and certain platforms, so it may not generalize well to more recent releases (scikit-learn developers, 2011).

References

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: With applications in R. Springer.
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